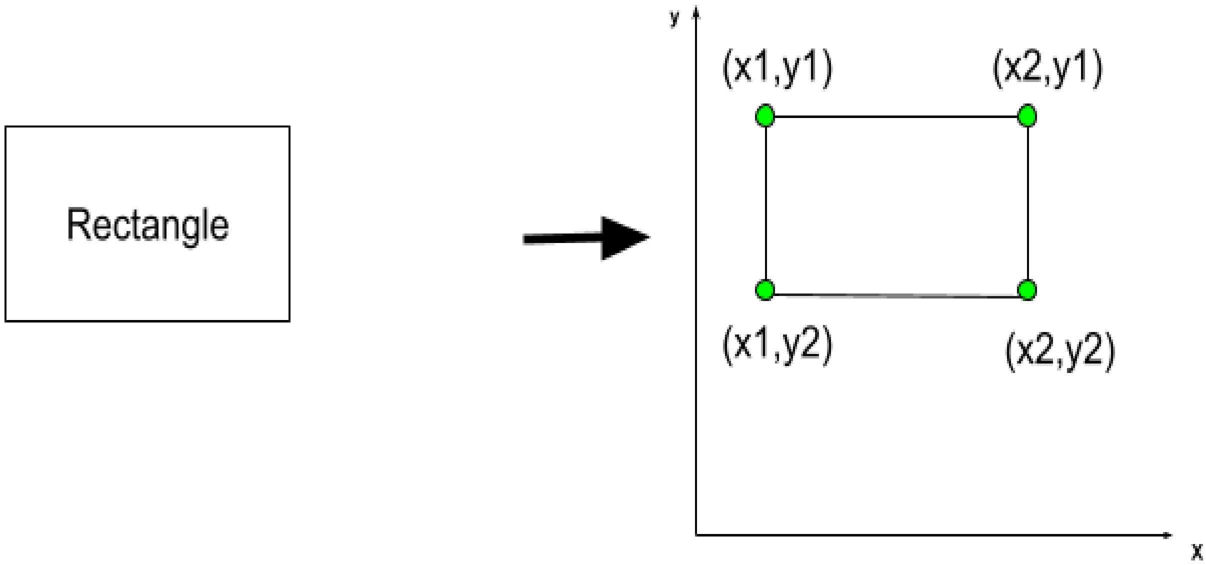
1.

a. Transform the rectangle problem into a problem on points

Each line contains four numbers x_1, y_1, x_2, y_2 representing the opposite corners of one axis-aligned rectangle.

From these its divided to (x_1,y_1), (x_1,y_2), (x_2,y_1), (x_2,y_2)

Like these it happen for all N rectangles, then it setup to 4N points.



b. The valid to consider the Convex hull of all rectangle corners as the optimal patrol path. Because its represent the smallest convex polygon that encloses all rectangles.

- Any valid enclosing path can only be shortened by "straightening" it.
- The path only needs to touch corner points.
- Rectangles that touch but don't overlap are safe.

c. The algorithm you will use to compute this convex hull (for example, Andrew's monotone chain or Graham scan), including:

- Sorting the points needed because of;
    - The monotone chain algorithm builds the hull from left to right.
    - Sorting guarantees a consistent traversal order.
    - It allows construction of:
        lower hull
        upper hull

- How the cross product is used
    - To determine whether three points make a left turn or right turn, use the cross product. Then we calculate the 2D cross product of vectors OA and OB:

      O (x_0,y_0) A(x_1,y_1) B(x_2,y_2)

      $(x_1 - x_0)(y_2 - y_1) - (y_1 - y_0)(x_2 - x_0)$

    - If cross > 0: left turn — A is on the convex side, keep it.
    - If cross ≤ 0: right turn or collinear — A is "inside" or redundant, remove it from the hull candidate stack

Then build the hull,

You do two passes over the sorted points: left→right to build the lower hull, then right→left to build the upper hull.

In each pass, you maintain a stack of hull candidates.

For each new point, you pop any point from the stack that makes a non-left turn (using the cross product test above), then push the new point.

- The overall time complexity in terms of N.
    - Sorting costs is O(N log N)
    - Each point is pushed and popped from the stack at most once, so both hull passes together cost O(N). The total is **O(N log N)**
    - An integer N with $1 \leq N \leq 2 \times 10^5$

2. **The cross product / orientation test of three points O, A, B.**

O (x_0,y_0) A(x_1,y_1) B(x_2,y_2)

$(x_1 - x_0)(y_2 - y_1) - (y_1 - y_0)(x_2 - x_0)$

Interpretation

If the value is > 0;

Then, the sequence O→A→B makes a counterclockwise turn (left turn).

If the value is < 0;

It makes a clockwise turn (right turn).

If the value is = 0;

then the points are collinear.

The Euclidean distance between two points.

P(x_1, y_1) Q(x_2,y_2)

Their Euclidean distance is:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This is giving the straight-line distance between two hull vertices. It is used when computing the perimeter of the convex hull.

The formula for the perimeter as the sum of distances along the convex hull.

Suppose the convex hull contains vertices:

$P_0, P_1, P_2, \ldots, P_{k-1}$

listed in counterclockwise order.

The perimeter is the sum of distances between consecutive vertices, including the edge returning to the starting point. The formula is;

$$perimeter = \sum_{i=0}^{n-1} \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2} + \sqrt{(x_0 - x_n)^2 + (y_0 - y_n)^2}$$

Where:

- (x_i, y_i) are the vertices of the convex hull in order (0 to n).
- (x_0, y_0) is the same as (x_{n+1}, y_{n+1}) to close the polygon loop.